

Bahdini Kurdish QA Generation

How 246,515 text chunks became 952,801 instruction-tuning records, and what is wrong with them

Corpus built 6–10 August 2026 · `google/gemini-3.1-flash-lite` · prompt v3 · \$301.41

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What this document is

This is the single source of truth for the QA-generation stage of the Bahdini Kurdish fine-tuning corpus. It begins where the chunking stage ended, with a finished work queue of 246,515 text chunks, and follows every step from there to the delivered dataset, including the exact prompt sent to the model, every token and dollar figure, the failure modes encountered, and the quality problems that remain unresolved. Design decisions are recorded with the reasoning behind them, so that a reader can tell which choices were forced, which were measured, and which are still open to revision.

This is the Markdown edition of the same report typeset in `qa_generation_report.tex`. Content is identical; the TikZ diagrams are rendered as Mermaid here.

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At a glance

INPUT		OUTPUT	
Chunks in the work queue	246,515	QA pairs generated	952,822
Source documents	4,306	Records delivered	952,801
Source corpora	8	Training tokens	644,306,453
Context tokens in queue	183,119,057	Pairs with reasoning	508,061
API CONSUMPTION		COST	
Calls made	248,243	Total spend	\$301.41
Input tokens billed	304,524,932	Per 1,000 pairs	\$0.316
Output tokens billed	150,188,299	Per source chunk	\$0.00122
Total tokens billed	454,713,231	Cost-model error	0.03%

The two token totals above measure different things and are not comparable. **454.7M** is what the generation provider billed: the prompt sent to Gemini plus the JSON it returned, counted by the provider's own tokenizer. **644.3M** is what the finished dataset contains: the assembled training records counted with Gemma's tokenizer. The delivered corpus is larger than the API traffic because each chunk's context is sent to the generator *once* but is then written into roughly three of the four training records built from it.

Where this report begins

The chunking stage is documented separately. This report takes its output as a given and starts there, but three properties of that output determine everything downstream and are restated here.

What a chunk is

A chunk is a contiguous passage of Bahdini text, packed paragraph-first to a target of 900 real Gemma tokens with a hard ceiling of 1,050 and a floor of 120. The target was not chosen for the generator's convenience: the partner specified a budget of roughly 1,000 tokens for the *prompt side* of a finished training record (system message plus question plus context, with the answer excluded), expressed as a mean rather than a cap. Measuring the fixed overhead of Gemma's chat template at 77 tokens for a representative record left approximately 900 tokens for context, which is where the target came from.

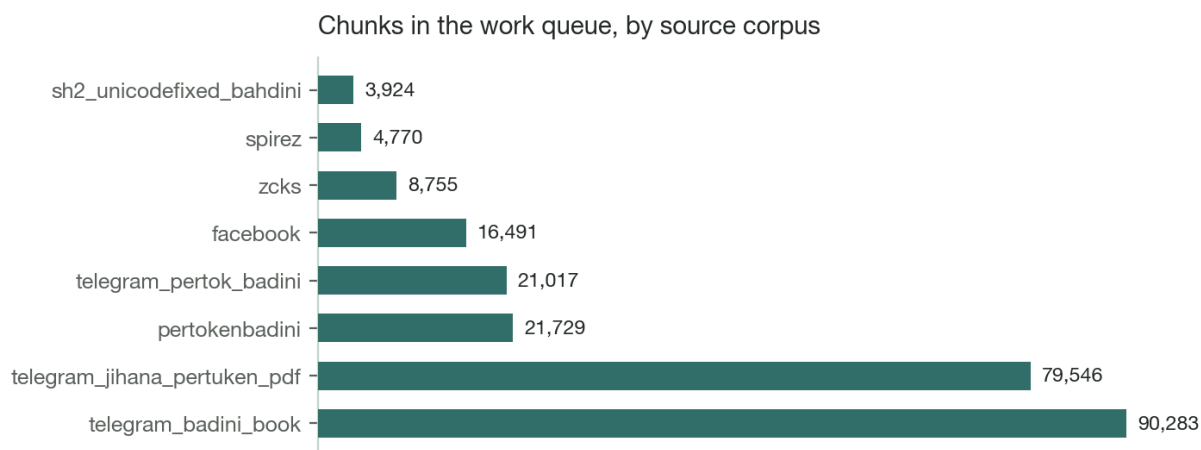
Chunk identifiers carry their origin

Every chunk id has the form `<origin>--<document_id>--<index>`. The origin prefix is load-bearing rather than decorative. The same source document can exist in both pools, native PDF text extraction and Gemini OCR of the same file, and both pools deliberately hash to the same `document_id` so that identifiers line up across pipelines. Before the prefix was added, 626 such documents produced colliding chunk ids: **41,457 duplicated ids spanning 82,914 chunks, 32.5% of the queue**. That collision would have silently corrupted both the resume logic and the chunk-text lookup at compile time. The current queue has zero collisions.

The queue survived two corruption gates

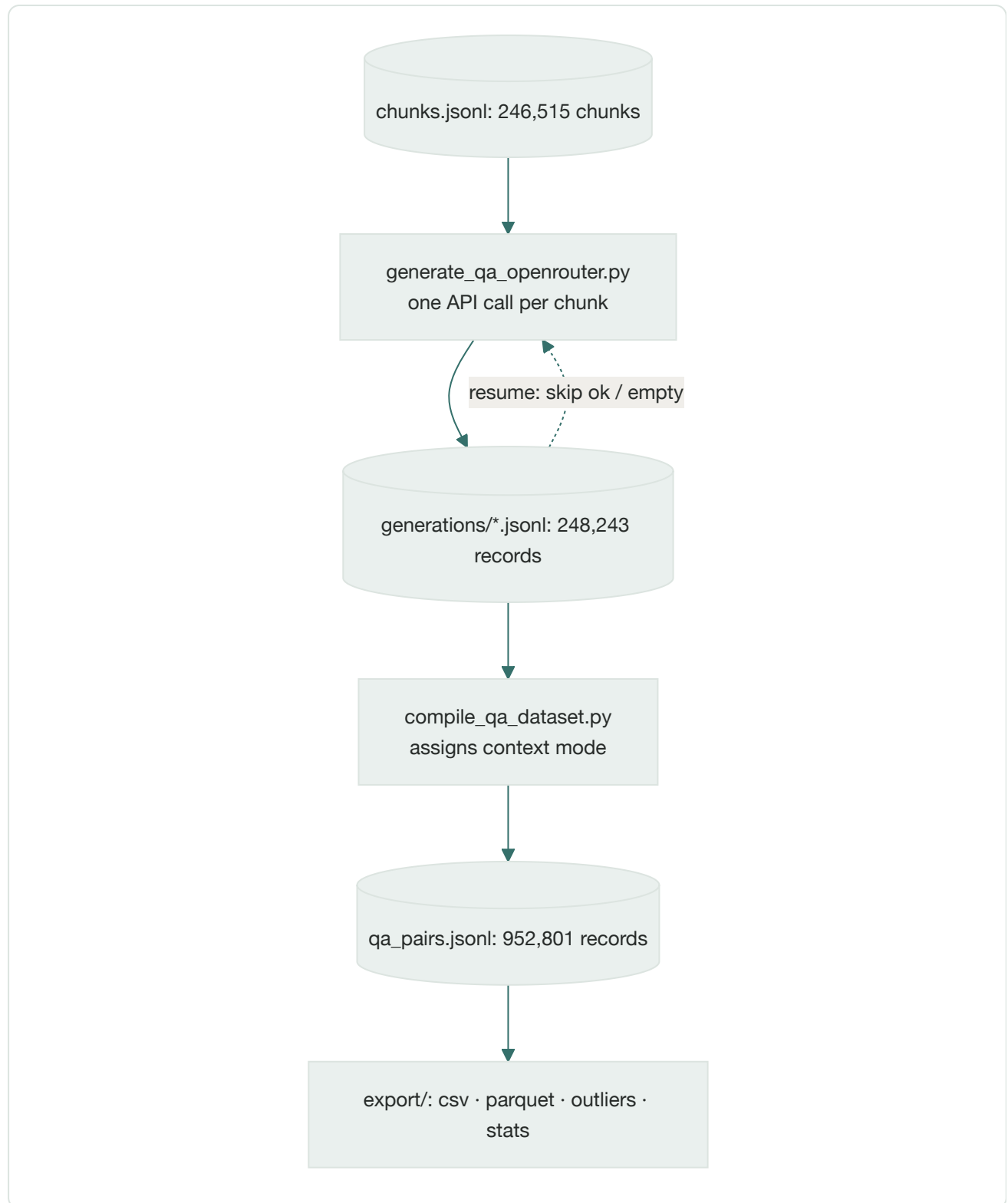
STAGE	COUNT	NOTE
Documents seen	5,219	after character-corruption backfill
Skipped, no usable chunks	16	shorter than the 120-token floor
Skipped, likely garbled	264	median chars/token below 1.5
Documents contributing chunks	4,939	
Chunks written	246,515	183,119,057 context tokens

The garbled-document gate deserves a note because it is the kind of check that is easy to get wrong. Legacy Kurdish font encodings can substitute the wrong Unicode characters wholesale, producing text with plausible Kurdish letter frequencies that spells no real words. Presentation-form ratios do not catch it. What separates it cleanly is the real tokenizer's chars-per-token ratio: verified-clean OCR text sits at 1.9–2.2, corrupted native extractions at 1.0–1.5. The gate is applied per document on the **median** across that document's own chunks, not per chunk, because line-wrapping noise makes individual chunks in a genuinely clean document dip as low as 0.12.



By origin the queue splits 166,169 chunks from the reviewed OCR corpus against 80,346 from native extraction, roughly two to one in favour of OCR, because OCR reads the rendered page image and is therefore immune to the PDF font problems that sent so many native extractions to the reject pile.

The pipeline end to end



The dashed edge is the resume path and is what made a four-day, restartable run possible. Each processed chunk appends one record to a per-document file, and a re-run reads those files first and skips anything already recorded as `ok` or `empty`. Failures are deliberately *not* recorded as done, so they are retried automatically on the next run without a separate pass.

Generation

The model, and why this one

PARAMETER	VALUE AND REASONING
Model	<code>google/gemini-3.1-flash-lite</code> via OpenRouter
Temperature	<code>0.7</code> : four pairs are requested in a single call, and lower temperatures made the four read as restatements of one another
Max output tokens	<code>2048</code> : four Bahdini pairs with reasoning average 605 output tokens, so this is headroom, not a target
Messages	<i>one</i> <code>user</code> message. No system role is sent
Response format	none: no JSON mode; the schema is enforced by the prompt and validated after the fact
Request timeout	180 seconds per attempt
Prompt version	<code>v3</code> , recorded on every output record

The tier was chosen by pilot rather than by assumption. Forty chunks were run at three pairs each and forty at four; `flash-lite` returned the full requested count on 39 of 40 either way, and the fourth pair read as a genuinely distinct question rather than filler. The stronger `pro` tier was therefore not needed, and the difference is roughly a factor of five in cost.

Two prompts, easily confused

The configuration defines `QA_GENERATION_PROMPT_TEMPLATE` and `QA_SYSTEM_PROMPT` , and they are unrelated. The first is what Gemini sees at generation time and never appears in the delivered dataset. The second is what Gemma sees at fine-tuning time, written into the `system` slot of finished records by the compile step, and never sent to Gemini. Editing the wrong one changes nothing about what is generated, or everything about the delivered records while leaving generation untouched.

The exact prompt

Reproduce this at any time with `python3 -c "import qa_config as c; print(c.build_qa_prompt('<CHUNK>'))"` . Rendered with `n_pairs=4` , the complete user message is:

You are building instruction-tuning data for a 100% pure Bahdini (Badini) Kurdish language model. Bahdini is spoken by people in Dohuk city and governorate.

Below is one excerpt ("the context") from a Bahdini text. Write 4 question-answer pairs a person could ask about this excerpt, as if using it for reading comprehension / instruction fine-tuning.

Rules:

- CRITICAL DIALECT RULE: Both the question and the answer MUST be written in 100% pure Bahdini Kurdish (Arabic script) exactly as spoken in Dohuk. Do NOT mix in any Sorani words, phrases, or grammar. Do NOT use general Kurmanji vocabulary or Latin script. You must ensure the dialect is strictly pure Bahdini.
- The answer must be grounded in the context: prefer extractive answers quoting or closely paraphrasing the text; reasonable inference/synthesis is fine for explanatory or summarization questions, but never introduce facts that are not supported by the context.
- Do not ask about the excerpt's formatting, page numbers, or the fact that it is an excerpt.
- Vary the question_type across the set; choose only from: factual, explanatory, summarization, definitional, inferential.
- If the context is too short, garbled, or content-free to support 4 distinct, well-grounded questions, return fewer pairs (an empty list is fine) rather than inventing filler.
- If answering the question genuinely requires connecting multiple details in the context or drawing a conclusion beyond a single stated fact, also fill in "reasoning": a brief step-by-step justification, in pure Bahdini Kurdish, for how the answer follows from the context. If the question is directly answerable by quoting or restating one explicit fact, set "reasoning" to null; do not pad it with filler.

Output strict JSON only: a list of objects, no markdown fences, no commentary before or after. Each object must have exactly these keys:

```
"question": string
"answer": string
"question_type": one of ['factual', 'explanatory', 'summarization', 'definitional', 'inferential']
"reasoning": string or null (see rule above)
```

Context:

```
""
{context}
""
```

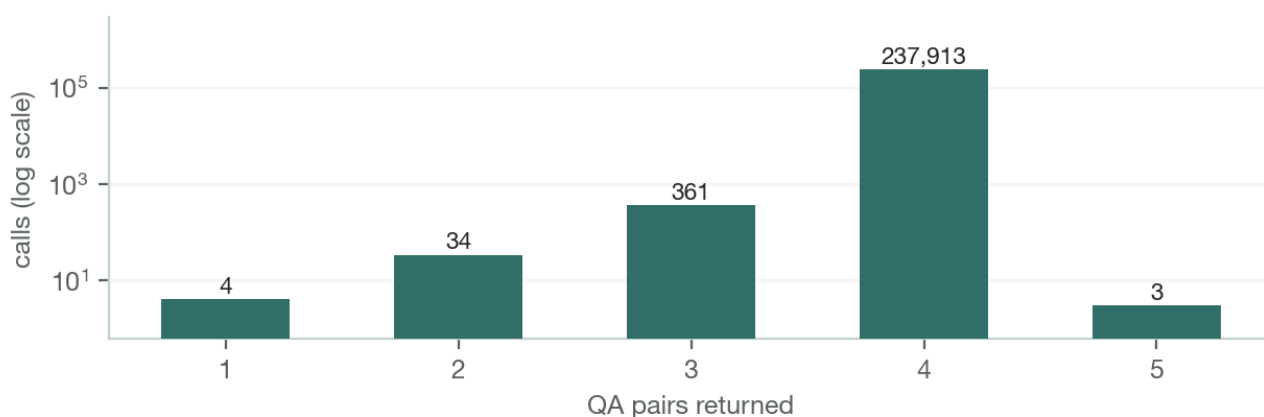
Two details worth knowing before anyone edits this. First, the `one of ['factual', ...]` line renders a raw Python list repr, quotes included, because that value is interpolated directly while the bullet above it is comma-joined. It is cosmetically untidy and the model complies with it regardless, but tidying it mid-corpus would mean a new prompt version, and a version bump part-way through leaves the delivered dataset generated under two different sets of instructions. Second, the shouty phrasing of the dialect rule is deliberate: version v2 said only "must be written in Bahdini Kurdish (do not switch to Sorani, Latin-script Kurdish, or Arabic)", and review found Sorani and general-Kurmanji vocabulary leaking in anyway. Version v3 names the three prohibitions separately and marks the rule critical.

Prompt version history

VERSION	CHANGE
v1	Initial. Three pairs per chunk, 700-token context target, no reasoning field.
v2	Added the optional <code>reasoning</code> field, the 70/30 context split, and the 900-token context target, after the partner confirmed the token budget covers the prompt side only.
v3	Current. Rewrote the dialect rule after Sorani leakage was observed; reasoning must also be pure Bahdini. All 248,243 records on disk are v3.

What the model returned

The model returned exactly 4 pairs on 99.8% of successful calls



Of 238,315 successful calls, 237,913 returned exactly the four pairs requested. **99.83%**. The parser accepts a variable count by design, so the handful that returned one, two, three or five pairs were kept rather than discarded. Five pairs is possible because the parser validates structure, not cardinality.

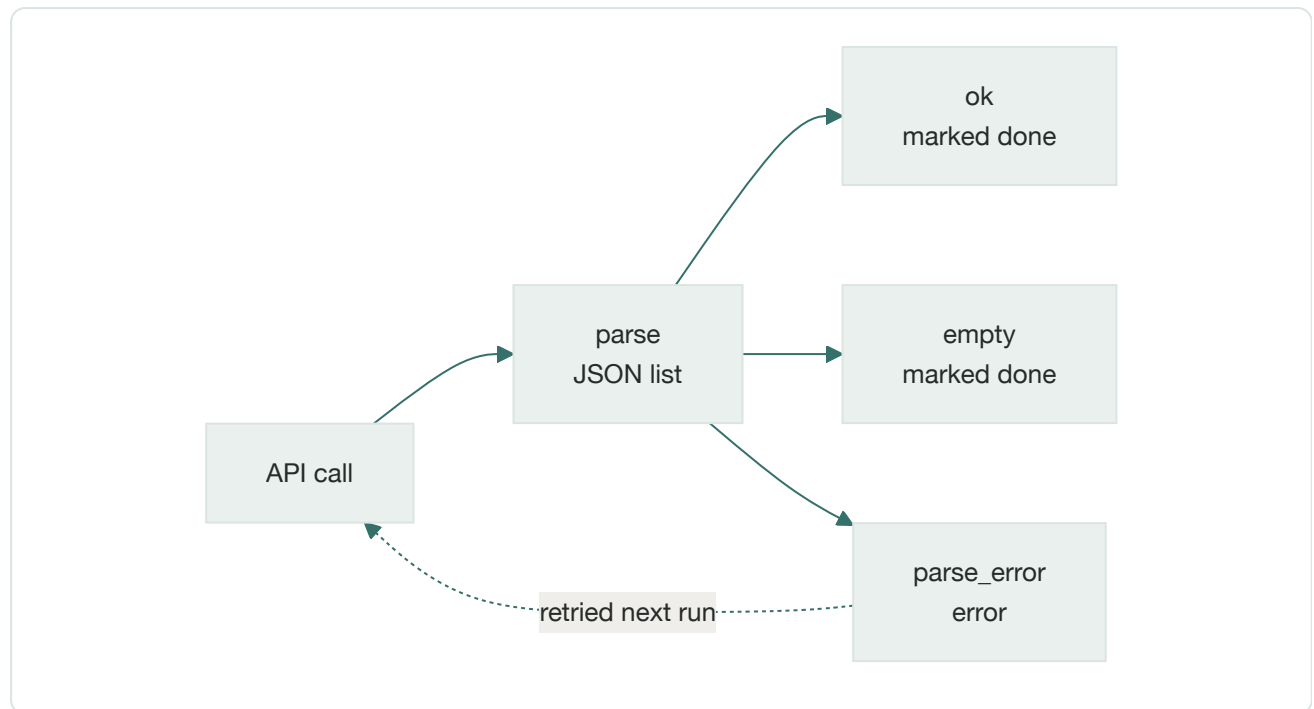
STATUS	CALLS	SHARE	MEANING AND RETRY BEHAVIOUR
<code>ok</code>	238,315	95.99%	at least one valid pair. Not retried
<code>empty</code>	8,209	3.31%	model returned <code>[]</code> : the intended answer for a garbled or content-free chunk. Not retried
<code>parse_error</code>	1,687	0.68%	response was not a JSON list, or every entry was malformed. Retried automatically
<code>error</code>	32	0.01%	HTTP or network failure after five attempts. Retried automatically

The empty responses are not failures. They are the model correctly declining to invent content for a chunk that cannot support four grounded questions, which the prompt explicitly instructs it to do. Treating that 3.31% as a defect would mean asking for filler. What genuinely produced nothing is `parse_error` plus `error`: 1,719 chunks, 0.70% of the queue.

Failure handling

Transport failures retry five times with exponential backoff plus jitter, computed as $2 \cdot 2^n + \text{rand}(0,1)$ seconds, on HTTP 429 and any 5xx. HTTP 402 and 403 are treated as terminal and are *not* retried: they mean credit exhausted or key rejected, and retrying them would burn five backoff cycles per chunk against a dead key while the run appeared to be working. Hitting either sets a stop flag that drains in-flight requests and ends the run cleanly.

At the pair level, the parser strips any markdown fence, parses the JSON, and keeps only entries with a non-empty question, a non-empty answer, and a `question_type` drawn from the five agreed values. A single malformed pair inside an otherwise-valid response is dropped; it does not fail the chunk.



Throughput, concurrency and the overnight run

The full corpus was generated across several sessions between 6 and 10 August 2026. Sustained throughput was roughly **450 chunks per minute at concurrency 32** with a dispatch batch of 128, up from about 230 per minute at concurrency 16.

Batch size matters more than it looks: the dispatcher waits for every request in a batch before starting the next, so the batch is a barrier. Setting it at or below the concurrency level starves the semaphore whenever a single slow request holds up its batch, which is why 128 pairs with a concurrency of 32 rather than the default 16.

The unattended portion ran under a supervisor that restarts the generator if it exits with work pending. That loop introduced one hazard worth recording: the generator's budget cap is evaluated *per run*, so a naive restart loop would silently reset the cap on every iteration. The supervisor therefore reads remaining credit back from the provider before each attempt and passes that, minus a small reserve, as that attempt's cap, so the ceiling tracks cumulative spend across restarts, with the provider's own key limit as a hard backstop underneath.

Cost, and a bug that mattered

QUANTITY	TOKENS	RATE (USD/M)
Input (prompt)	304,524,932	0.25
Output (completion)	150,188,299	1.50
Total billed	454,713,231	

Mean consumption per call was 1,227 input and 605 output tokens. Total spend was **\$301.41**: \$0.316 per thousand pairs, or \$0.00122 per source chunk.

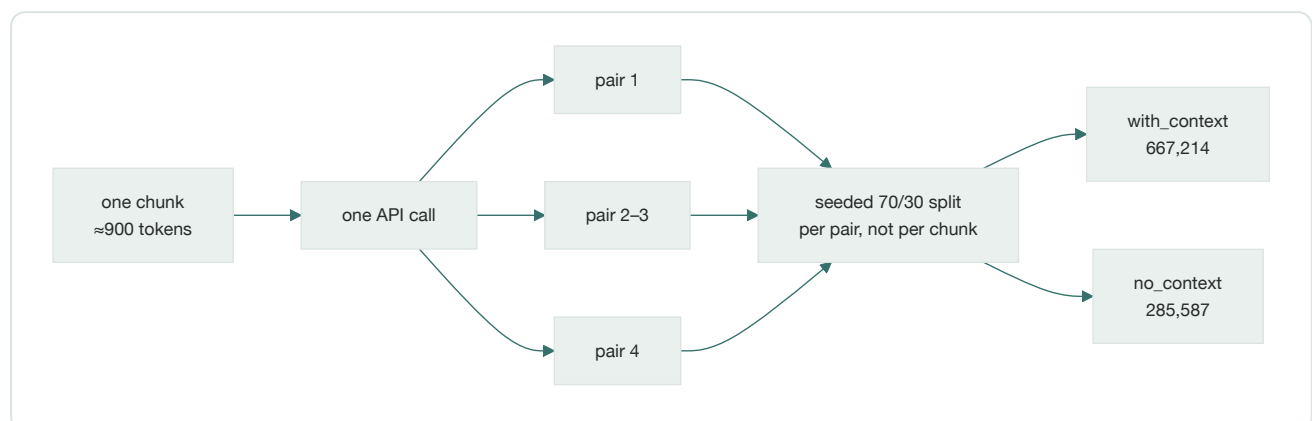
Pricing is keyed by exact model slug. It previously was not, a single pair of constants for the `pro` tier was applied to every run regardless of which model was actually called, which overstated `flash-lite` runs by a factor of 3.75. That was not merely a display error. **The budget cap is evaluated against this estimate**, so an inflated rate makes a run halt far short of its real budget while appearing to have completed: the first full-corpus attempt carried a \$350 cap and would have stopped at roughly 32% of the queue having actually spent about \$93.

The cost model is validated, not assumed

Recomputing every record on disk from its stored token counts gives \$50.04 for the first API key; the provider's own usage endpoint reports \$50.0542 for the same key. That is a 0.03% discrepancy across 41,467 calls, which is why the figures in this report are stated as measurements rather than estimates. An unknown model slug deliberately falls back to the most expensive known rate, so the failure mode is stopping early rather than overspending.

From pairs to training records

How 246,515 chunks became 952,801 records



The context mode is assigned **per pair** with a fixed seed, not per chunk. That matters: assigning per chunk would mean all four pairs from a passage share a mode, which correlates the split with document content. A fixed seed makes the assignment reproducible across re-runs of the compile step over the same generation records rather than reshuffled each time.

The delivered count is 952,801 rather than the 952,822 pairs on record. The difference is 21 pairs referencing six chunk ids that no longer exist in the queue, orphaned when the corpus was rebuilt after the character-corruption fix. They are dropped rather than delivered without their grounding text.

Record anatomy

```
{
  "messages": [
    {"role": "system", "content": "Answer the question in Bahdini Kurdish using the supplied context."},
    {"role": "user", "content": "Context: <chunk>\n\nQuestion: <question>"},
    {"role": "assistant", "content": "<answer>", "reasoning": "<present only when needed>"}
  ],
  "metadata": {
    "document_id": "<source document id>",
    "chunk_id": "<source chunk id>",
    "source": "<corpus name>",
    "question_type": "<one of five>",
    "context_mode": "with_context | no_context"
  }
}
```

For a `no_context` record the user message is only `Question: <question>`, and the system prompt switches to "Answer the question in Bahdini Kurdish", dropping "using the supplied context", which would be false when there is none. That second system string was a judgement call rather than a specified requirement, and is a one-line change if different wording is preferred.

The reasoning field is native, not invented

Gemma 4's own chat template reads `message.get('reasoning')` directly off the assistant message and renders it into the model's thought channel ahead of the answer. Verified directly against the template: passing `reasoning` as a sibling key of `content` produces a `<|channel>thought` block, whereas inlining a hand-rolled marker into the content itself is treated as literal answer text. This is why reasoning lives in its own field rather than being concatenated into the answer.

The finished dataset

Size in tokens

Every record was tokenized with the real `google/gemma-4-31B-it` tokenizer. The chat template's own per-record overhead was measured at 18.68 tokens on a sample rather than assumed.

Where the 644.3M training tokens sit

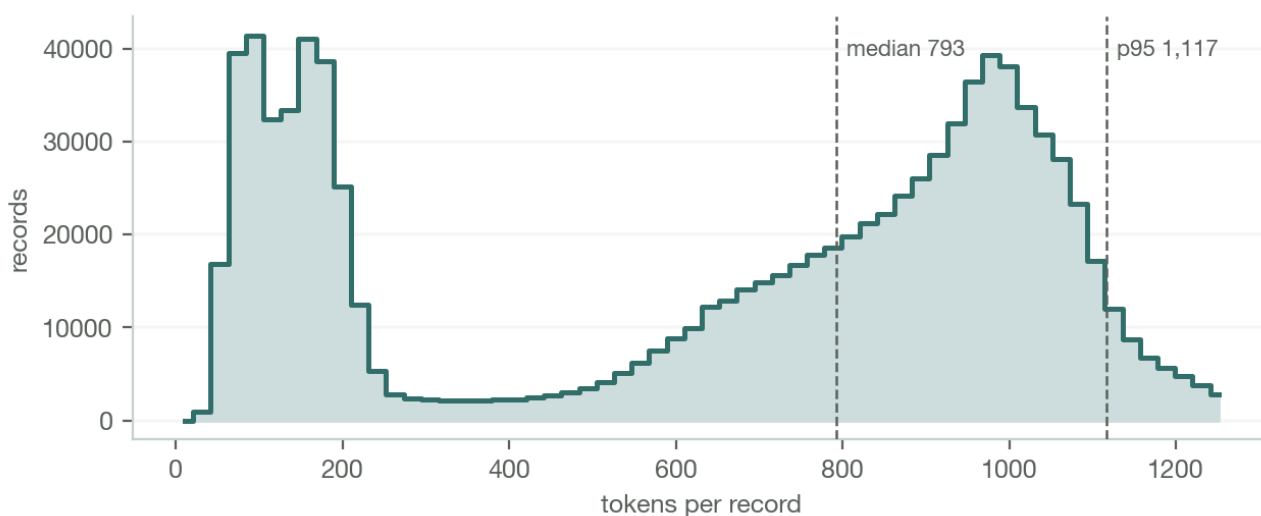


COMPONENT	TOKENS	SHARE
User message (context + question)	526,428,585	81.7%
Assistant answer	56,938,845	8.8%
Reasoning	31,896,635	5.0%
Chat template overhead	17,798,323	2.8%
System prompt	11,244,065	1.7%
Total	644,306,453	100%

The single most consequential number here is the first one. **Context is 82% of the training budget**, so any proposal to trim, filter or re-balance context is a proposal about almost the entire dataset, while any change to answers or reasoning touches at most a seventh of it.

Length distribution

Record length is bimodal, not centred on its mean



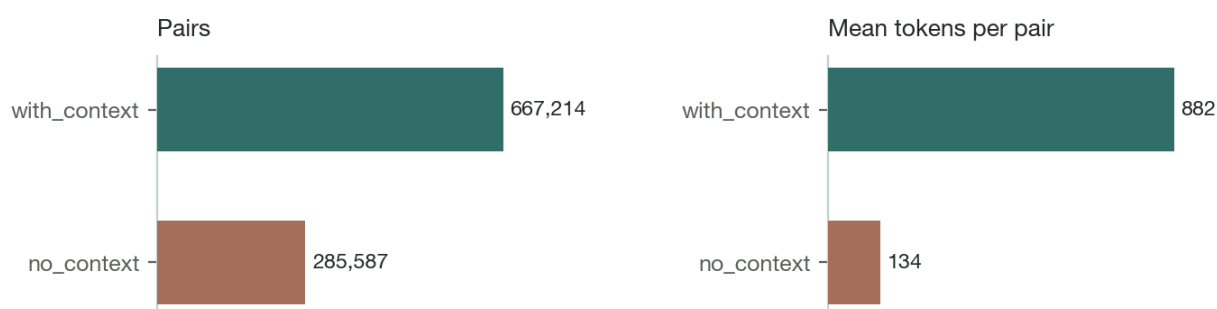
The mean is the wrong summary for this dataset. Record length is bimodal: bare-question records cluster near 90 tokens and context-carrying ones near 950, and the mean of 658 falls in the valley between them, describing almost no actual record. The two populations should be reported, and batched, separately.

FIELD	MIN	P05	MEDIAN	MEAN	P95	MAX
User (prompt side)	3	23	691	553	992	1,340
Assistant (answer)	2	24	59	60	100	399
Whole record	26	80	793	658	1,117	1,535

Against the partner's roughly 1,000-token prompt-side mean, measured p95 is 992 and only 27 records in the entire corpus exceed the 1,300-token flag threshold. a 30% allowance over the target. The budget was met comfortably.

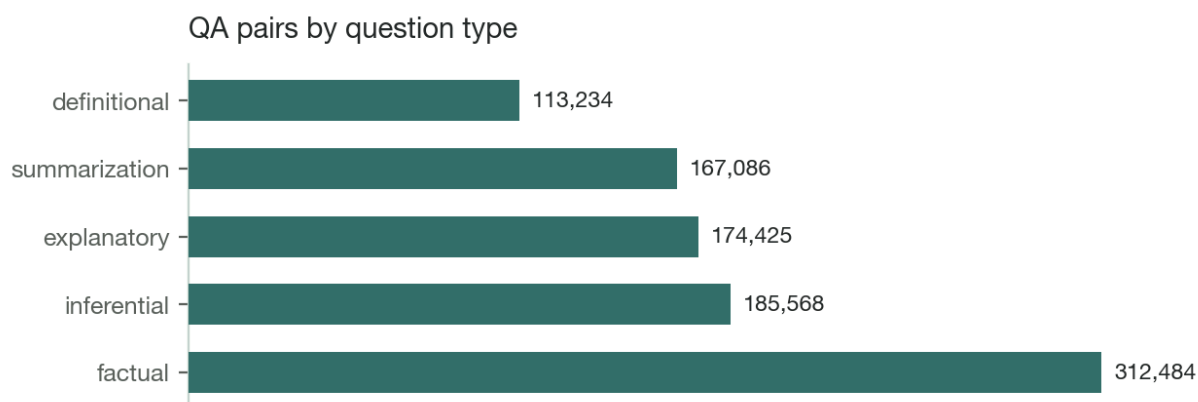
Context modes

The 70/30 context split, and what it costs in tokens



The realised split is 667,214 `with_context` against 285,587 `no_context` : **70.03%** / **29.97%**, matching the configured ratio of 0.7 to two decimal places. The two modes exist because the model will be served both ways (retrieval-augmented and bare), and a model trained only on context-bearing prompts tends to refuse or hedge when the context is absent. Both modes were generated with full context visible to the generator, so the bare-question pairs remain grounded in real source text even though that text is not shown at training time.

Composition



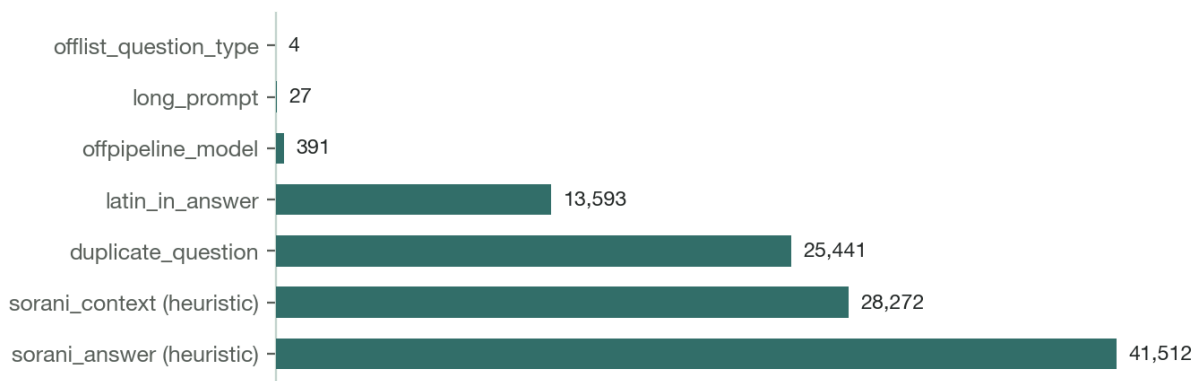
SOURCE CORPUS	PAIRS	SHARE
telegram_badini_book	350,085	36.7%
telegram_jihana_pertuken_pdf	307,525	32.3%
pertokenbadini	85,524	9.0%
telegram_pertok_badini	81,012	8.5%
facebook	60,442	6.3%
zcks	33,971	3.6%
spirez	18,728	2.0%
sh2_unicodefixed_bahdini	15,514	1.6%

Two Telegram book collections are 69% of the corpus. This reflects what Bahdini text exists in bulk rather than any sampling decision, but it has a direct consequence: **any train/eval split must be stratified by source**, or the evaluation set measures those two collections and nothing else.

Quality findings

Every anomaly found is collected in `qa_outliers.csv`, one row per flagged pair, keyed by `row_index` back into the delivered file so any decision maps to specific records. A pair can carry several flags.

Flagged pairs by issue, 103,357 rows in total



FLAG	ROWS	NATURE
sorani_answer	41,512	heuristic
sorani_context	28,272	heuristic; the case that leaks into training
duplicate_question	25,441	objective
latin_in_answer	13,593	objective; violates the prompt outright
offpipeline_model	391	objective; from a one-off experiment
long_prompt	27	objective; over the 1,300-token flag
offlist_question_type	4	objective
Total flagged rows	103,357	10.85% of the dataset

5,750 pairs carry two or more flags. Those are the highest-value rows to read first, because independent signals agreeing is far stronger evidence than any single heuristic firing alone.

The two dialect flags are a queue, not a verdict

They come from a marker-frequency heuristic with no validated threshold, and several of its markers occur in both dialects. The counts indicate how much material is worth inspecting; they do not state how many rows are wrong. Both `sorani_score` and `bahdini_score` are exposed as columns precisely so the cutoff can be re-judged against real labels. Every other flag in the table above is an objective, checkable property.

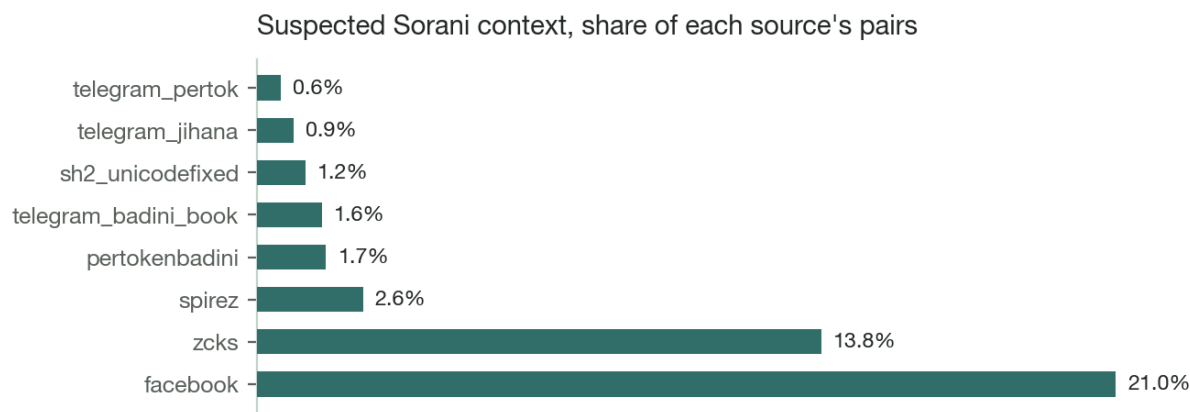
Sorani source text

The most consequential finding. Some source chunks are Sorani rather than Bahdini. The generator handles this correctly, it obeys the dialect rule and writes the question and answer in Bahdini regardless. The leak is at compile time: for `with_context` records the raw chunk is placed in the user message, so Sorani text enters the training prompt of an otherwise-Bahdini record.

Sorani context found in the facebook corpus, chunk text, excerpt

قهلهمه پارکه رهکهی باوکم ... بۆ وا به کول دمهگریت؟

Sorani orthography throughout, the `هکهی` definite suffix and the `دهگریت` verb form are not Bahdini. The question and answer generated from this chunk are correct Bahdini; the context beneath them is not.



Plotted as a rate rather than a count deliberately: by absolute count the large Telegram corpora appear worst simply because they contain 350,000 pairs each. Normalising shows the problem is concentrated in `facebook` (21.0%) and `zcks` (13.8%), an order of magnitude above the book corpora, which is consistent with provenance, since those two are social posts and a mixed-dialect website rather than curated Bahdini books. The same concentration appeared in an independent estimate made on a 2,404-row stratified sample when the corpus was only 17% complete, which is weak corroboration that the signal is real even though the threshold is not calibrated.

Latin-script leakage

13,593 answers (1.43%) contain at least one Latin character, against a prompt that forbids Latin script outright. The rate was also 1.43% when the corpus was 17% complete, so this is structural rather than an artefact of the later run. The class is mixed: proper nouns and citations alongside genuine leakage. The tail is unambiguous, the worst single record carries 244 Latin characters, which turn out to be Emily Dickinson's "*I never saw a moor*" sitting in the source text.

Duplicate questions

25,441 pairs (2.7%) repeat a question string seen earlier. **Do not deduplicate on the question alone.** The repeats are dominated by generic questions, the kind askable of almost any passage, which arise independently across unrelated chunks and carry entirely different answers. Global deduplication on the question string would delete legitimately distinct training pairs. Deduplicate on the `(question, answer)` pair, or on question within a document.

Off-schema rows

175 chunks, producing 391 pairs, were generated by a one-off manual experiment using a different model outside this pipeline. They carry no token counts and therefore contribute nothing to any cost figure in this report. They are also the only source of the four off-list `question_type` values in the corpus (`descriptive` three times, `comparative` once), which the pipeline's own parser would have rejected. The pairs are otherwise valid Bahdini and were left in place; the compile step does not re-validate `question_type`, so filter there if strict schema conformance is required.

Open decisions

None of the following are settled. Each requires a Bahdini speaker's judgement, and each has an inexpensive fix once decided.

1. **Calibrate the dialect threshold.** Label a few hundred rows from the `facebook` and `zcks` slice by eye, then set the cutoff against those labels. Everything else on this list is cheaper than this, and none of it is blocked by it.
2. **Handle Sorani contexts.** The cheap fix is at compile time: force affected pairs to `no_context` rather than dropping them, which keeps the Bahdini question and answer and discards only the Sorani prompt text. The thorough fix is a per-document dialect gate in the chunking stage, applied the same way as the existing corruption gate, per document, because dialect is a property of the document and a per-chunk cutoff would shred documents on line-wrapping noise.
3. **Triage Latin leakage.** Sort by `latin_chars_in_answer` descending. The tail is obviously wrong and the head is mostly proper nouns; a threshold somewhere between will do once someone reads twenty rows.
4. **Deduplicate carefully.** On `(question, answer)` or within a document. never on the question string globally.
5. **Stratify the evaluation split by source**, or it will measure two Telegram corpora and nothing else.
6. **Decide on the 395 off-schema pairs.** Small enough to simply drop if strict conformance matters.

Artifacts and reproduction

FILE	SIZE	PURPOSE
<code>qa_pairs.jsonl</code>	2.4 GB	the deliverable, 952,801 records
<code>parquet/</code> (4 shards)	350 MB	same rows, Hub-ready, verified identical
<code>qa_pairs.csv</code>	2.2 GB	same rows, spreadsheet encoding
<code>qa_outliers.csv</code>	251 MB	103,357 flagged pairs, 8 flags
<code>stats.json</code>	18 KB	token counts and distributions
<code>sample.jsonl</code>	77 KB	40 records, every type × mode combination

The three encodings of the deliverable are not approximations of one another. The table exporter reconstructs records from the written table and diffs them against the source JSONL; the current result is 952,801 against 952,801 with zero mismatches.

Rebuilding from scratch

```
python3 qa_generation/pipeline/build_chunks.py
python3 qa_generation/pipeline/generate_qa_openrouter.py --concurrency 32 --batch-size 128
python3 qa_generation/pipeline/compile_qa_dataset.py --sample-size 40
python3 qa_generation/export/compute_dataset_stats.py
python3 qa_generation/export/export_dataset_table.py --verify
python3 qa_generation/export/export_outliers.py
```

Re-running the generator is always safe and always makes progress: completed chunks are skipped, failures are re-attempted, and interruption at any point loses nothing because every finished chunk is already flushed to disk.

Figures generated from `stats.json` and `qa_outliers.csv`. Token counts from `google/gemma-4-31B-it`.
prepared by: sawab.aziz@newrozholdings.com